

# A Historical Analysis of the Vienna Physics Curricula

Dating the Topics of the Bachelor's and Master's Programmes in Physics (Version 2026)

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## Abstract

This report documents an analysis of the curricula of the Bachelor's and Master's programmes in Physics at the University of Vienna (both Version 2026). For each curriculum the individual scientific topics of all modules were extracted from the official curriculum documents and decomposed into atomic items. Each topic was then annotated with the year in which the corresponding concept was first introduced or the corresponding discovery was made, together with the responsible scientist(s) and a bibliographic source for the attribution. The resulting datasets—545 topics for the Bachelor's programme and 174 for the Master's programme—were analysed statistically and visualised as histograms of topics per decade. The Bachelor's curriculum spans more than four centuries of physics and mathematics (median year 1905), whereas the Master's curriculum is sharply concentrated in the twentieth century (median year 1936), reflecting its role as a research-oriented specialisation built on the broad foundations laid in the Bachelor's degree.

## 1 Objective

The goal of this analysis was to characterise the two Physics curricula not by their administrative structure but by the *historical depth* of their scientific content. Concretely, for every topic taught we asked: *when*, and *by whom*, was this piece of physics or mathematics first established? Aggregating the answers reveals how each programme is distributed across the history of science and makes the pedagogical difference between the Bachelor's and Master's degrees quantitatively visible.

## 2 Data and Method

### 2.1 Source material

The analysis is based on the two official curriculum documents, `BA_Physik.pdf` (Bachelor) and `MA_Physics.pdf` (Master), both in the Version 2026 draft. The text of every module description was extracted and the enumerated content (the “*Die Inhalte umfassen*” / possible-content lists) was used as the basis for the topic catalogue. Learning-outcome and competence statements were deliberately excluded, as they describe skills rather than datable scientific concepts.

### 2.2 Topic decomposition

The curriculum texts frequently bundle several distinct concepts into a single phrase. Every such phrase was decomposed into *atomic* topics, each corresponding to a single discovery or concept that can carry one year and one attribution. For example, the Bachelor entry “*Gradient, Divergenz, Rotation*” was split into three separate rows, and the large elective modules *Wahl-Fachbereiche* and *Wahlaborpraktikum* were unfolded into their constituent named methods (e.g. FFT, quantum error correction, BCS superconductivity). The elective subject areas were retained

as an explicit grouping column so that the topics of each field (Computational Physics, Quantum Information, Particle Physics, ...) remain identifiable.

## 2.3 Annotation

Each atomic topic received four annotations:

- **Year** — the year of the seminal publication that first introduced the concept or reported the discovery.
- **Scientist(s)** — the originator(s); where a topic combines several contributions, the principal one is named with secondary contributors noted in parentheses.
- **Source** — a bibliographic reference for the attribution, given as *author, landmark work, year*. Publication titles are kept in their original language, which is standard scholarly practice.
- **Subject area** (electives only) — the field heading under which the topic is offered.

Topics without a single identifiable originator—laboratory and presentation skills, orientation courses, generic programming setup, the master’s thesis and defence, and similarly broad method headings—were marked “—” and excluded from the statistical analysis.

## 2.4 Deliverables and reproducibility

For each curriculum the catalogue was produced as a spreadsheet in both German and English (identical content, translated headers, module names, subject areas and topics). The decade histograms are generated directly from the spreadsheets by two short, self-contained Python scripts (`make_histograms.py` and `make_combined_histogram.py`), so the figures can be regenerated at any time. The complete list of files is given in Section 7.

## 2.5 Caveats

The attributions are, by their nature, scholarly judgements. For foundational results with multiple independent originators, and especially for the broad elective and core subject areas that represent entire research fields, a single *representative* seminal contribution was chosen. The source column documents the work that established each concept to the level of *author–title–year*; exact volume and page numbers were intentionally omitted to avoid spurious precision. Individual attributions can therefore be revised without affecting the overall picture.

# 3 The Topic Catalogues

## 3.1 Bachelor’s programme

The Bachelor catalogue contains 545 atomic topics across 25 modules, of which 476 carry a year and 69 are non-attributable. The two large elective modules dominate in size: *Wahl-Fachbereiche* contributes 149 topics across eleven subject areas and *Wahllaborpraktikum* a further 62 across eight. The core lecture modules (Experimental Physics, Theoretical Physics, Analysis, Linear Algebra, ...) contribute the remainder.

## 3.2 Master’s programme

The Master catalogue contains 174 atomic topics across 8 modules, of which 140 carry a year and 34 are non-attributable. Here the scientific content is concentrated almost entirely in a single module: *Core Topics* provides 150 topics, organised into 15 subject areas (Atmospheric Aerosol

Physics, Biophysics, Computational Physics, Condensed Matter Physics, Electronic Structure, Experimental Quantum Physics, General Relativity and Cosmology, Quantum Field Theory, Phase Transitions and Critical Phenomena, Physics of Nuclei and Isotopes, Non-equilibrium Statistical Physics, Scattering/Microscopy/Spectroscopy, Soft Matter Physics, Theoretical Particle Physics, and Theory in Quantum Optics and Quantum Information). The remaining modules (Scientific Research Methods, Advanced Research Topics, Electives, Proposal/Master’s Seminar, Thesis and Defensio) are research- and skills-oriented and contribute few datable topics.

## 4 Results

### 4.1 Summary statistics

Table 1 summarises the distribution of topic years for the two programmes. For the Bachelor’s programme the single ancient entry (Euclidean geometry,  $\approx 300$  BC) is treated as an outlier and excluded from the summary figures; it is the only pre-Renaissance topic and, when included, shifts the mean down to 1877 and inflates the standard deviation from 82 to 129 years.

Table 1: Summary statistics of the topic years (dated topics only). Bachelor figures exclude the single Euclidean-geometry outlier.

| Quantity               | Bachelor        | Master       |
|------------------------|-----------------|--------------|
| Dated topics $n$       | 475             | 140          |
| Median year            | 1905            | 1936         |
| Mean year              | 1882            | 1936         |
| Std. deviation (years) | 82              | 35           |
| Earliest topic         | 1572 (Bombelli) | 1828 (Green) |
| Latest topic           | 2004 (graphene) | 2001 (KLM)   |
| Modal decade           | 1920s (51)      | 1930s (22)   |

### 4.2 Bachelor’s programme

The distribution of Bachelor topics by decade is shown in Figure 1. The curriculum covers the entire history of modern physics and its mathematical underpinnings: a sparse seventeenth- and eighteenth-century tail (Newton, Leibniz, Euler, Lagrange), a pronounced rise through the nineteenth century (classical electrodynamics, thermodynamics, and the rigorous foundations of analysis), and a dominant early-twentieth-century peak in the **1920s** (51 topics) driven by the quantum-mechanical revolution. Content continues at a substantial level up to the present day through quantum information, condensed-matter and computational topics. The oldest non-ancient topic is the introduction of complex numbers (Bombelli, 1572); the most recent is graphene (Geim and Novoselov, 2004).

### 4.3 Master’s programme

The corresponding distribution for the Master’s programme is shown in Figure 2. It is markedly more modern and far more tightly concentrated: there is essentially no content before  $\sim 1820$ , and the mass of the distribution lies in the twentieth century with a peak in the **1930s** (22 topics). The median year is 1936 and the standard deviation is only 35 years—less than half that of the Bachelor’s programme. The earliest topic is Green’s functions (Green, 1828); the most recent is photonic quantum information processing (Knill, Laflamme and Milburn, 2001).

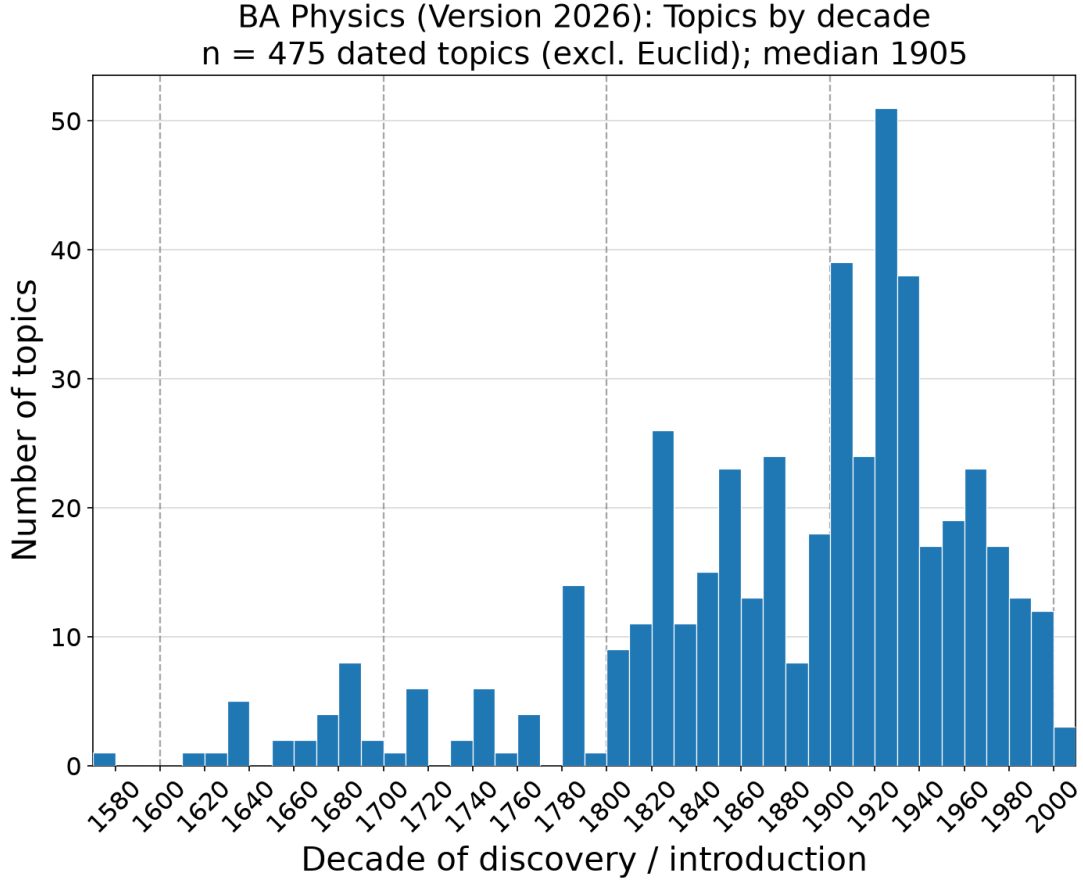


Figure 1: Number of Bachelor topics per decade. Vertical dashed lines mark centuries. The distribution peaks in the 1920s and has a median year of 1905.

#### 4.4 Comparison

Figure 3 overlays the two distributions on a common axis. The contrast captures the pedagogical logic of the two-cycle degree structure: the Master’s distribution (orange) sits almost entirely beneath the modern end of the Bachelor’s distribution (blue). The Bachelor’s programme builds the historical and mathematical *foundations* of physics across four centuries, while the Master’s programme assumes those foundations and specialises in twentieth- and twenty-first-century *research* physics. Numerically, the centre of gravity moves forward by about three decades, from a median of 1905 to 1936.

## 5 Discussion

The quantitative results confirm and sharpen an intuitive expectation. A Bachelor’s curriculum in physics must, of necessity, recapitulate the historical development of the field: one cannot teach quantum mechanics without first teaching classical mechanics, calculus, and electrodynamics. This produces a broad distribution with a long tail reaching back to the seventeenth century (and, for Euclidean geometry, to antiquity). A research-oriented Master’s curriculum, by contrast, can take this entire edifice as given and devote itself to the modern research frontier; its topic distribution is therefore both later in time and far narrower. The roughly 30-year forward shift of the median, and the halving of the temporal spread, are compact quantitative signatures of this transition from *foundational breadth* to *research-level depth*.

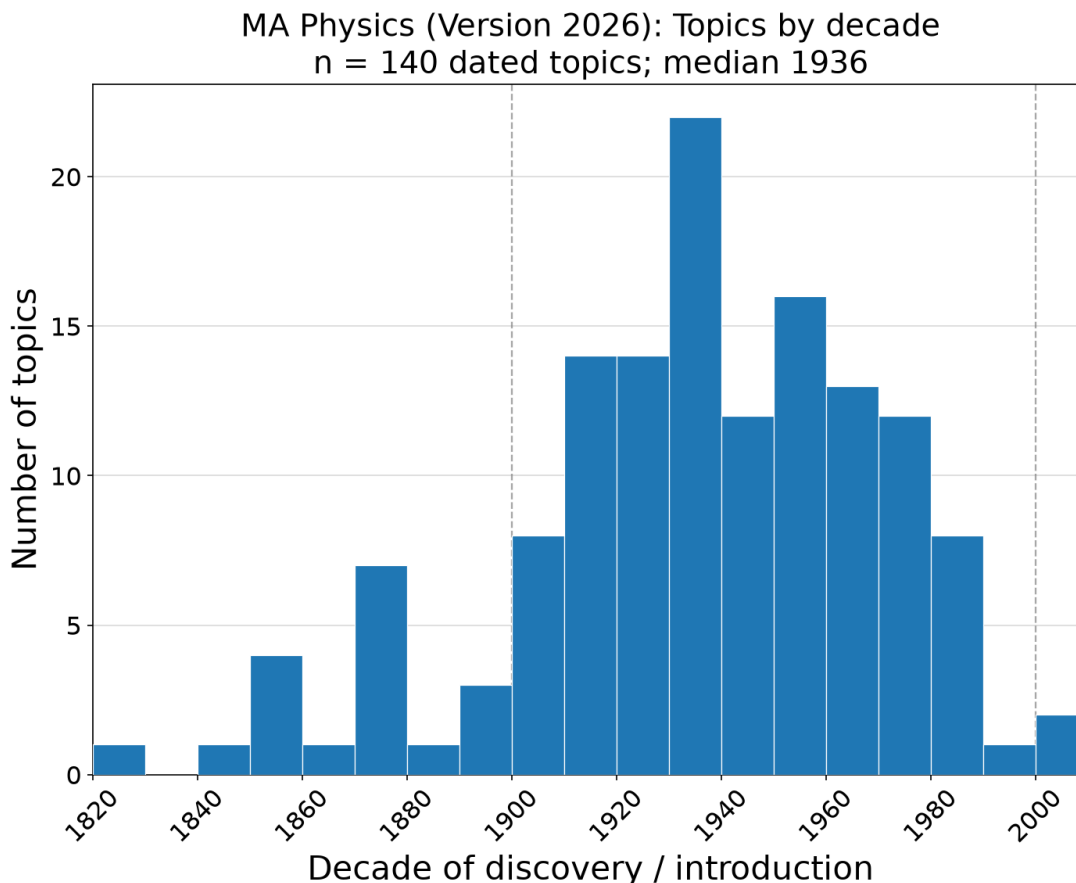


Figure 2: Number of Master topics per decade. The distribution is concentrated in the twentieth century with a median year of 1936.

## 6 Limitations

**Earliest-date bias.** The most important limitation concerns the meaning of the assigned year. By construction, each topic is dated by the *earliest* possible date—the year in which the concept was first introduced or the founding discovery was made—whereas a course actually treats the topic together with all of its subsequent developments. The year therefore marks the *birth* of a subject, not the full historical range of the material taught under that heading.

A clear example is the topic *Molecular dynamics* in the Computational Physics course, which is dated to 1957, the year of the first hard-sphere molecular-dynamics simulation by Alder and Wainwright. The course, however, does not stop there: it also covers much more recent methodological developments such as thermostat and barostat algorithms for sampling different ensembles (1980s), constraint algorithms, and symplectic / time-reversible integration schemes, many of which date from the 1980s and later. All of this modern content is nevertheless represented in the dataset by the single founding year 1957.

The same effect applies throughout: a topic such as “density functional theory” (dated 1965) is taught together with exchange–correlation functionals and methods developed decades later, and “general relativity” (dated 1915) is accompanied by modern results on gravitational waves and numerical relativity. As a consequence, the histograms *systematically shift weight towards earlier decades* and should be read as the distribution of the *conceptual origins* of the curriculum, not as the distribution of the actual age of the material presented in the lecture halls. The true content is, on average, more modern than the founding-year distributions suggest. A more faithful but considerably more laborious alternative would be to assign each topic a range of

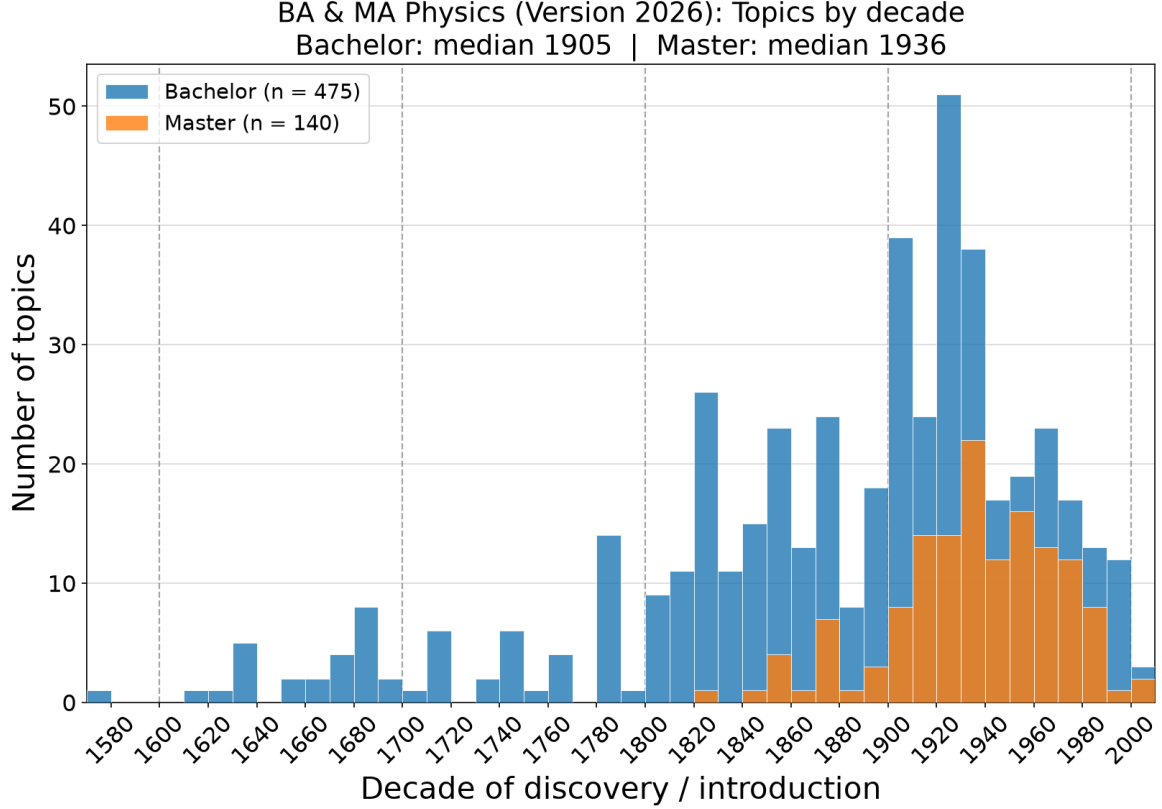


Figure 3: Combined histogram of Bachelor (blue) and Master (orange) topics by decade. The Master’s content is confined to the modern end of the Bachelor’s range. Counts are raw; the Master’s programme has fewer topics in total (140 vs. 475 dated topics).

years, or to record separately the principal later milestones; this was beyond the scope of the present analysis.

**Further limitations.** Three additional points should be kept in mind. First, the year and scientist assignments are scholarly judgements; for fields rather than single discoveries a representative seminal work was chosen. Second, the histograms weight every atomic topic equally, irrespective of the teaching time (ECTS) devoted to it, so the analysis measures the *conceptual* rather than the *temporal* balance of the curriculum. Third, the decomposition of compound topics into atomic items necessarily involves some discretion, which affects the absolute counts (but not the qualitative shape of the distributions).

## 7 Files Produced

The analysis produced the following deliverables, all in the project directory:

- **Spreadsheets (Bachelor):** BA\_Physik\_Module\_Themen.xlsx (German), BA\_Physics\_Modules\_Topics.xlsx (English).
- **Spreadsheets (Master):** MA\_Physik\_Module\_Themen.xlsx (German), MA\_Physics\_Modules\_Topics.xlsx (English).
- **Histograms (Bachelor):** BA\_Physik\_Themen\_Histogramm.png (German), BA\_Physics\_Topics\_Histogram.png (English).

- **Histograms (Master):** MA\_Physik\_Themen\_Histogramm.png (German), MA\_Physics\_Topics\_Histogram.png (English).
- **Combined histograms:** BA\_MA\_Themen\_Histogramm.png (German), BA\_MA\_Topics\_Histogram.png (English).
- **Scripts:** make\_histograms.py, make\_combined\_histogram.py.

Each spreadsheet contains a *Topics by Module* sheet (one row per atomic topic, with module, subject area, topic, year, scientist and source) and a *Module Overview* sheet (topic counts per module).